

Persona Mixtures for Predicting Human Strategic Behavior

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Replication Project

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Background emphasis: Appendix A.2 “Constructing the sample of AI agents” in *General Social Agents* (arXiv:2508.17407v5)¹

¹<https://arxiv.org/html/2508.17407v5#A2>

What problem is the paper solving? (high-level)

- We want an AI agent population whose **action distribution** matches humans across settings.
- “Out-of-the-box” LLM behavior can be systematically biased (e.g., always favoring one option).
- Key idea: use **theory-grounded prompt templates** and **optimize parameters / mixtures** on seed data, then validate on distinct-but-related settings.

Appendix A.2 example: construct dictator-game agents from *efficiency*, *self-interest*, *inequity aversion* and optimize on multiple dictator variants simultaneously.²

²<https://arxiv.org/html/2508.17407v5#A2>

Appendix A.2 “mean idea” in one sentence

Elicit $\Rightarrow$ Optimize $\Rightarrow$ Validate

- **Elicit:** encode *theory parameters* directly into prompts
 - e.g., efficiency / self-interest / inequity aversion levels
 - **Optimize:** search (grid / Bayesian optimization) over parameters and/or a small persona mixture
 - pick θ^* or weights w^* minimizing a divergence to human data on seed games
 - **Validate:** evaluate out-of-sample on *new but related* games (holdout settings)
 - report generalization metrics (e.g., KL, MAE, calibration)
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- **Theory grounding** reduces search space and improves generalization.
 - **Validation across distinct settings** prevents overfitting to a single dataset.

Appendix A.2 instantiates this for allocation (dictator) games.³

³<https://arxiv.org/html/2508.17407v5#A2>

Why this matters: Simulating humans with LLMs in social science

- Social science theories aim to **predict behavior across settings**, but applying them to novel contexts often requires ad-hoc tweaks.
- LLMs offer a low-cost, scalable alternative: prompt-based agents that draw on vast pre-training knowledge + explicit instructions.
- Goal: Build **“general” agents** that match human action distributions in *new* games/settings without game-specific data.
- Applications: policy design, mechanism testing, behavioral experiments in silico, multi-agent simulations of markets/societies.

The promise (Manning & Horton, 2025)

“A general, low-cost method for accurately simulating human behavior with AI agents would have wide application in the social sciences.”

Emerging literature: Argyle et al. (2022), Binz & Schulz (2023), Park et al. (2023), Hewitt et al. (2024), and many others show LLM agents can approximate humans — but results are inconsistent across domains.

Literature: LLM agents predicting human behavior in games

- **Early successes:** LLMs simulate humans in novel studies (Hewitt et al., 2024; Binz et al., 2025; Suh et al., 2025).
- **Strategic games:** Level-k reasoning, cognitive hierarchy models approximated via prompts (Manning & Horton, 2025; Lorè et al., 2024).
- **Persona-based approaches:** Demographic/personality prompts shift behavior (e.g., Big Five, MBTI) — but often stylistic, not behavioral (Park et al., 2023; PersonaLLM, 2023/2025 updates).
- **Mixtures & optimization:** Mixture-of-Personas (MoP) models subpopulations (2025); theory-grounded trait optimization (Manning & Horton Appendix A.2).

Related: Xie et al. (2025) use LLMs to categorize games/motivations; multi-agent dynamics in dilemmas (2025–2026 papers); but generalization remains fragile.

Key insight from Manning & Horton (2025)

Theory-grounded prompts + multi-setting optimization + out-of-sample validation $\gg$ atheoretical / stylistic personas.

Why predicting humans with LLMs is hard

- **Overfitting risk:** Arbitrary prompts can match one dataset perfectly but fail catastrophically elsewhere (e.g., “offer \$6–\$9” in dictator game).
- **Underfitting with baselines:** Off-the-shelf LLMs often biased (e.g., always egalitarian, overly cooperative, or deterministic).
- **Theory vs. style:** Stylistic personas (MBTI, historical figures) collapse to near-identical kernels $\Rightarrow$ unidentifiable mixtures, poor generalization.
- **Validation difficulty:** Single-dataset train/test split cannot detect overfitting; needs *distinct but related* holdout settings.
- **Model/setting sensitivity:** Behavior varies across LLMs, temperatures, contexts, even languages.

Core challenge (Manning & Horton, 2025)

Achieving *in-sample* fit is easy; building agents that *generalize reliably* to novel but related settings is hard — requires theory grounding + careful optimization.

Do behavioral (Strategic) vs. stylistic (MBTI/Historical) personas differ in generalization?

Level-k reasoning: Core model of bounded sophistication

Level-k models (Nagel 1995; Stahl & Wilson 1994/95) explain limited strategic thinking:

- People do **not** perform infinite levels of reasoning (as in Nash equilibrium).
- Instead, they use a finite, heterogeneous number of reasoning steps.

Level-k hierarchy (iterated best response):

- **Level-0 (L0):** Non-strategic / naïve players
 - Choose randomly, pick salient numbers, or follow a simple heuristic (often around 50 in $[0,100]$ games).
- **Level-1 (L1):** Best-responds to Level-0
 - Assumes everyone else is L0 $\rightarrow$ best response to that belief.
- **Level-2 (L2):** Best-responds to Level-1
 - Assumes opponents are L1 $\rightarrow$ best response to L1 behavior.
- And so on: Level-k assumes opponents are Level-(k-1).

Most people stop at L1 or L2 in one-shot games; L3+ is rare.

Level-k mechanism: How it works in the p-beauty contest (Nagel 1995)

Game setup ($p = 2/3$):

- Players simultaneously choose $x \in [0, 100]$.
- Winner(s): closest to target $= \frac{2}{3} \times$ (average of all choices).
- Nash equilibrium: everyone chooses 0 (iterated deletion of dominated strategies).

Step-by-step level-k reasoning mechanism:

- **L0**: Picks a salient number (e.g., 50, midpoint of $[0, 100]$).
- **L1**: Assumes opponents are L0 $\rightarrow$ expects average $= 50 \rightarrow$ chooses $\frac{2}{3} \times 50 \approx 33.3$.
- **L2**: Assumes opponents are L1 $\rightarrow$ expects average $= 33.3 \rightarrow$ chooses $\frac{2}{3} \times 33.3 \approx 22.2$.
- **L3**: Assumes opponents are L2 $\rightarrow$ expects average $= 22.2 \rightarrow$ chooses $\frac{2}{3} \times 22.2 \approx 14.8$.
- **L4**: ≈ 9.9 , and so on $\rightarrow$ converges toward 0.

Each additional level of reasoning halves the previous guess (roughly) $\rightarrow$ creates characteristic spikes in the distribution.

Level-k in human data: Empirical patterns ($p=2/3$ beauty contest)

Observed human behavior (Nagel 1995 and many replications):

- Clear spikes / modes around:
 - ~ 33 (L1 players)
 - ~ 22 (L2 players)
 - ~ 15 (L3 players)
- Mean choice typically 35–40 (mixture dominated by L1 and L2).
- Very few choices near 0 (almost no high-level reasoning in one-shot play).
- Average reasoning depth: ~ 1 –2 steps.

Key empirical regularities:

- L1 and L2 are modal across many subject pools (students, experts, newspaper readers).
- Depth is limited and heterogeneous; experience/learning shifts choices downward.
- Cognitive hierarchy (CH) variant (Camerer et al. 2004) often fits better: players best-respond to a *mixture* of lower levels.

→ These spikes provide strong evidence for bounded rationality $\Rightarrow$ excellent target for theory-grounded LLM prompting (e.g., Strategic personas).

Level-k reasoning: Key empirical patterns from human data

Summary of human results (survey: Crawford et al. 2013)

- L1 and L2 are the most common (modal) levels in one-shot strategic games:
 - p-beauty contest / guessing games
 - 11–20 money request game
 - Dictator-game variants with strategic elements
- Reasoning depth is limited and **inconsistent** across games and individuals (Cooper 2024).
- **Cognitive hierarchy (CH)** variant (Camerer et al. 2004) often fits human data better:
 - Players best-respond to a *Poisson-distributed mixture* of lower levels (not assuming everyone is exactly $L(k-1)$).

Overall: Humans exhibit bounded, heterogeneous sophistication — rarely exceeding 2–3 steps of reasoning in one-shot play.

The 11–20 game (Arad & Rubinstein 2012) & implications for LLMs

11–20 money request game (Arad & Rubinstein 2012) — clean level-k structure:

- Two players simultaneously request an integer amount $x \in \{11, 12, \dots, 20\}$.
- Each receives their requested amount.
- Bonus rule: If request is *exactly one less* than opponent's, you get +20 (undercut incentive).

Level-k predictions (natural anchors):

- L0: Requests 20 (max payoff, no strategic thinking).
- L1: Requests 19 (best response assuming opponent picks 20).
- L2: Requests 18 (best response assuming opponent picks 19).
- L3: Requests 17, and so on.

Implication for LLM agents: Explicit level-k style prompts (e.g., “Think one step ahead of naive opponents” or “Assume others pick randomly and best-respond”) capture this bounded rationality $\Rightarrow$ produce **distinct behavioral kernels** $\Rightarrow$ enable effective mixtures and better out-of-sample generalization (unlike purely stylistic personas).

$\rightarrow$ Aligns with Appendix A.2: theory-grounded traits outperform generic/style-based approaches.

From bounded rationality evidence to theory-grounded LLM agents

What the level-k literature (and related games) tells us:

- Humans exhibit **bounded, structured strategic reasoning** (L0–L2 dominant, clear best-response hierarchies).
- These patterns are **consistent enough** across games and populations to be useful modeling targets.
- Stylistic / surface-level personas (MBTI, historical figures) rarely reproduce such structured behavioral differences.

Key opportunity for LLM agents:

- Explicitly **encode behavioral theories** into prompts (level-k depth, self-interest strength, fairness concern, etc.)
- $\Rightarrow$ create **distinct, interpretable action kernels**
- $\Rightarrow$ enable meaningful mixtures that generalize beyond training data

$\rightarrow$ Theory-grounded prompting should outperform generic or stylistic approaches — exactly the hypothesis tested in Appendix A.2 of Manning & Horton (2025).

Appendix A.2: Putting theory-grounded prompting into practice

Manning & Horton (2025, Appendix A.2) apply this philosophy to “dictator / allocation games”:

- Choose **three core behavioral traits** from established economic theory:
 - Efficiency concern
 - Self-interest / own-payoff maximization
 - Inequity aversion / fairness preference
- **Parameterize** each trait on an interpretable scale (1–10)
- **Optimize** trait combinations on multiple related training games simultaneously
- **Validate** generalization on held-out but conceptually similar settings

Why this is powerful

Theory-grounded parameterization + multi-setting fitting + careful validation

⇒ avoids overfitting to idiosyncrasies of any single game

⇒ produces agents that generalize meaningfully to new (but related) environments

We now look at how they operationalize this approach.

Appendix A.2: Parameterized prompt template (concept)

A.2 constructs agents by embedding **three interpretable traits** into the prompt:

$$\theta = (\theta_{\text{eff}}, \theta_{\text{self}}, \theta_{\text{ineq}}) \in \{1, \dots, 10\}^3,$$

where higher values mean stronger preference for:

Efficiency : maximize total payoffs,

Self-interest : maximize own payoff,

Inequity aversion : prefer fairness / equality.

Interpretation

Each θ defines a distinct agent prompt $\pi(\theta)$; sampling multiple θ yields a **candidate agent set**.

Source: A.2 prompt parameterization.⁴

⁴<https://arxiv.org/html/2508.17407v5#A2>

Appendix A.2: Why multiple training settings?

Let $\mathcal{G}_{\text{fit}} = \{g_1, \dots, g_S\}$ be multiple dictator variants. Then optimizing on

$$\sum_{s=1}^S d(P_{\text{hum}}(\cdot | g_s), P_{\text{agent}}(\cdot | g_s))$$

reduces the chance of “cheating” by fitting idiosyncrasies of one game.

Paper's message

Theory grounding + multi-setting optimization + out-of-sample validation $\Rightarrow$ robust predictive gains.

Appendix A notes two technical differences: multi-setting optimization and parameterized prompt template.⁵

⁵<https://arxiv.org/html/2508.17407v5#A2>

Objects and notation (our replication aligns with mixture view)

- Games: $g \in \mathcal{G}set$; actions: $a \in \mathcal{A}_g$ (discrete).
- Human empirical distribution: $P_{\text{hum}}(a \mid g)$.
- Persona-conditioned LLM distribution: $P_k(a \mid g)$ for persona $k \in \mathcal{K}$.

Mixture model

$$P_{\text{mix}}(a \mid g; w) = \sum_{k=1}^K w_k P_k(a \mid g), \quad w \in \Delta_K := \left\{ w \in \mathbb{R}_+^K : \sum_k w_k = 1 \right\}.$$

Training objective: KL minimization

Fit weights on seed games $\mathcal{G}_{\text{fit}}$:

$$\hat{w} \in \arg \min_{w \in \Delta_K} \sum_{g \in \mathcal{G}_{\text{fit}}} \text{KL}(P_{\text{hum}}(\cdot | g) \| P_{\text{mix}}(\cdot | g; w)),$$

where

$$\text{KL}(P \| Q) = \sum_{a \in \mathcal{A}_g} P(a) \log \frac{P(a)}{Q(a)}.$$

Relative improvement (reported)

$$\text{Rel_Improve} = 1 - \frac{\sum_{g \in \mathcal{G}_{\text{eval}}} \text{KL}(P_{\text{hum}}^g \| P_{\text{mix}}^g)}{\sum_{g \in \mathcal{G}_{\text{eval}}} \text{KL}(P_{\text{hum}}^g \| P_{\text{base}}^g)}$$

When optimization becomes ill-posed: persona collapse

If persona kernels are (nearly) identical,

$$P_1(\cdot \mid g) \approx P_2(\cdot \mid g) \approx \dots \approx P_K(\cdot \mid g) \quad \forall g,$$

then for any $w, w' \in \Delta_K$,

$$P_{\text{mix}}(\cdot \mid g; w) \approx P_{\text{mix}}(\cdot \mid g; w'),$$

so w is weakly identified and the optimizer can stagnate at uniform weights.

Diagnostic (collapse score)

$$\text{Collapse}(P) = \max_{a \in \mathcal{A}_g} P(a), \quad (\text{close to } 1 \Rightarrow \text{near-deterministic}).$$

Algorithm 1: End-to-end persona mixture pipeline

Algorithm 1 Persona-mixture fitting and evaluation

Require: Personas $\{\pi_k\}_{k=1}^K$, games $\mathcal{G}_{\text{fit}}, \mathcal{G}_{\text{eval}}$, sampling sizes $N_{\text{fit}}, N_{\text{eval}}$

1: **Estimate persona kernels on seed games:**

2: **for** $g \in \mathcal{G}_{\text{fit}}$ **do**

3: **for** $k = 1, \dots, K$ **do**

4: Sample N_{fit} LLM plays under persona π_k ; extract actions

5: $\hat{P}_k(\cdot | g) \leftarrow$ empirical frequencies

6: **end for**

7: **end for**

8: **Fit weights:**

$$\hat{w} \leftarrow \arg \min_{w \in \Delta_K} \sum_{g \in \mathcal{G}_{\text{fit}}} \text{KL}(P_{\text{hum}}^g \| \sum_k w_k \hat{P}_k^g)$$

9: **Evaluate on holdout games:**

10: **for** $g \in \mathcal{G}_{\text{eval}}$ **do**

11: **for** $k = 1, \dots, K$ **do**

12: Sample N_{eval} plays; form $\hat{P}_k(\cdot | g)$

13: **end for**

14: $\hat{P}_{\text{mix}}^g \leftarrow \sum_k \hat{w}_k \hat{P}_k^g$

15: Record $\text{KL}(P_{\text{hum}}^g \| \hat{P}_{\text{mix}}^g)$ and compare to baseline

16: **end for**

Algorithm 2: Simple exponentiated-gradient on the simplex (optional)

Let

$$J(w) = \sum_{g \in \mathcal{G}_{\text{fit}}} \text{KL} \left(P_{\text{hum}}^g \parallel \sum_k w_k P_k^g \right).$$

Exponentiated gradient (mirror descent) update:

$$w_k^{(t+1)} \propto w_k^{(t)} \exp \left(-\eta \frac{\partial J(w^{(t)})}{\partial w_k} \right), \quad \sum_k w_k^{(t+1)} = 1.$$

Interpretation

Multiplicative updates preserve positivity and naturally respect simplex constraints.

Effective number of personas (sparsity summary)

Define inverse Simpson index:

$$N_{\text{eff}}(w) = \frac{1}{\sum_{k=1}^K w_k^2}.$$

- $N_{\text{eff}} = 1$: one persona dominates (degenerate mixture).
- $N_{\text{eff}} = K$: uniform mixture.

Example (Strategic top-3)

If $w = (0.69, 0.28, 0.03)$ (others ≈ 0):

$$N_{\text{eff}} \approx \frac{1}{0.69^2 + 0.28^2 + 0.03^2} \approx 1.79.$$

Replication setup (What I ran)

- Seed and evaluation sampling: $N_{\text{fit}} = 50$, $N_{\text{eval}} = 100$; master seed = 7.
- Game families: BASIC, COSTLESS, CYCLE.
- Persona families tested: **Strategic**, **MBTI**, **Historical**.
- Valid extraction: `valid_rate=1.000` for mix and baseline across runs.

Baselines

Single-prompt (“base”) distribution compared against fitted mixture.

Our replication: Strategic family (KL improves)

Family	KL(hum mix)	KL(hum base)	Rel. improve	valid_rate
BASIC	13.4264	14.6342	+8.3%	1.000
COSTLESS	4.2162	5.9841	+29.5%	1.000
CYCLE	2.1803	3.2115	+32.1%	1.000

Fitted sparsity

Top-3 mix weights: (20, 0.69), (18, 0.28), (19, 0.03).

Our replication: MBTI family (often no gain; collapse)

MBTI	Rel. improve	KL(hum mix)	KL(hum base)
BASIC	-20.1%	17.5751	14.6342
COSTLESS	-8.9%	6.5142	5.9841
CYCLE	+30.3%	2.2379	3.2115

Symptom

Estimated w remained uniform across 16 types; BASIC action distribution collapsed: Top-3 mix: (18, 1.0), (11, 0), (12, 0).

Our replication: Historical family (strong negative; KL blows up)

Historical	Rel. improve	KL(hum mix)	KL(hum base)
BASIC	-20.1%	17.5751	14.6342
COSTLESS	-138.0%	14.2422	5.9841
CYCLE	-344.3%	14.2703	3.2115

Mechanism (KL sensitivity)

If $P_{\text{mix}}(a) \approx 0$ for actions with $P_{\text{hum}}(a) > 0$, KL can explode. Uniform weights across 18 historical personas + near-identical kernels $\Rightarrow$ weak identification + poor support.

Paper vs. replication: what matches the Appendix A.2 worldview?

A.2 emphasizes theory-grounded traits

A.2 explicitly parameterizes prompts by interpretable behavioral traits (efficiency/self-interest/inequity aversion) and searches parameters to match human data across multiple training settings.^a

^a<https://arxiv.org/html/2508.17407v5#A2>

- Our **Strategic** personas behave like “behavioral traits” $\Rightarrow$ nontrivial mixture and KL gains.
- **MBTI/Historical** are largely stylistic $\Rightarrow$ kernels nearly identical, weights unidentifiable, frequent collapse.

Takeaway aligned with A.2

Behavioral grounding **matters more than** persona flavor/style.

Conclusion

- We formalize human behavior prediction as matching **action distributions** via convex mixtures of persona-conditioned LLM policies.
- Fitting w by minimizing KL on seed games improves out-of-sample accuracy **when personas induce distinct behavioral kernels**.
- Strategic personas: consistent KL reductions (up to $\sim 32\%$).
- MBTI/Historical personas: frequent **collapse / near-identical kernels** $\Rightarrow$ little gain or large degradation (KL sensitivity).

How this connects back to Appendix A.2

A.2's recipe (parameterized, theory-grounded prompts + multi-setting optimization + validation) provides a principled way to avoid “style-only” personas and improve generalization.^a

^a<https://arxiv.org/html/2508.17407v5#A2>

Backup: Optional slide for A.2 workflow (BO on parameters)

Appendix A.2 (allocation games) uses Bayesian optimization over trait parameters $\theta \in \{1, \dots, 10\}^3$ and evaluates each candidate by repeated sampling per setting, then selects the best parameter vectors and forms a small mixture of agents.⁶

Abstracted meta-algorithm

$$\theta \sim \text{search} \Rightarrow \hat{P}_\theta(\cdot \mid g) \Rightarrow \text{score}(\theta) = \sum_{g \in \mathcal{G}_{\text{fit}}} d(P_{\text{hum}}^g, \hat{P}_\theta^g) \Rightarrow \theta^*$$

- Their distance metric in Appendix A is MAE over Left rates (binary actions).
- Our implementation uses KL over discrete action distributions (multi-action games).

⁶<https://arxiv.org/html/2508.17407v5#A2>